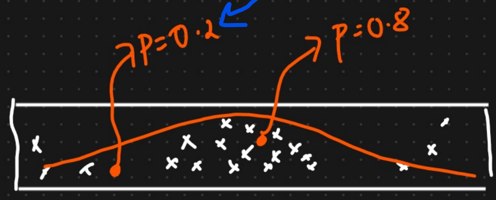
P value

The p value is a number, calculated from a statistical test, that describes how likely you are to have found a particular set of observations if the null hypothesis were true. P values are used in hypothesis testing to help decide whether to reject the null hypothesis.



Space bar

Out of 100 touches, we touch around 20 times in this region

Hypothesis Testing

Eg: Coin is Fair or Not {100 times}

① Null Hypothesis:

Ho: Coin is fair

$p(H) = 0.5$ $p(T) = 0.5$ consider as fair

$p(H): 0.6$ $p(T) = 0.4$ consider as fair

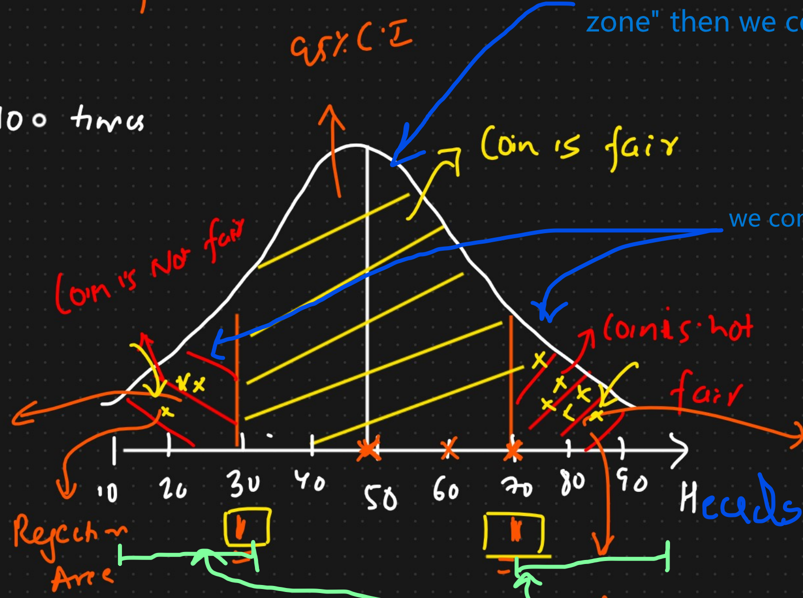
$p(H) = 0.7$ $p(T) = 0.3$ considering as not fair

② Alternate Hypothesis:

H_1: Coin is not fair

③ Experiment : 100 times

If the value fall in this area "confidence zone" then we consider coin is fair coin



Otherwise

we consider coin as not fair

100 times tossing coin

How do we define confidence interval for that we use significance value

Rejection Area

④ Significance Value : $\alpha = 0.05$

confidence interval $C.I = 1 - 0.05 = 0.95$

⑤ Conclusion $P <$ Significance value

P > significance value

Reject the Null Hypothesis

else

$\boxed{\text{Fail to Reject}}$ the Null Hypothesis